

Relative Binding Free-Energy Calculations at Lipid-Exposed Sites: Deciphering Hot Spots

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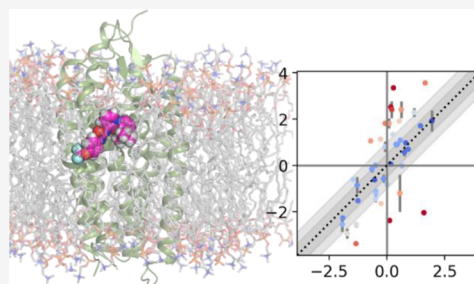


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ABSTRACT: Relative binding free-energy (RBFE) calculations are experiencing resurgence in the computer-aided drug design of novel small molecules due to performance gains allowed by cutting-edge molecular mechanic force fields and computer hardware. Application of RBFE to soluble proteins is becoming a routine, while recent studies outline necessary steps to successfully apply RBFE at the orthosteric site of membrane-embedded G-protein-coupled receptors (GPCRs). In this work, we apply RBFE to a congeneric series of antagonists that bind to a lipid-exposed, extra-helical site of the P2Y1 receptor. We find promising performance of RBFE, such that it may be applied in a predictive manner on drug discovery programs targeting lipid-exposed sites. Further, by the application of the microkinetic model, binding at a lipid-exposed site can be split into (1) membrane partitioning of the drug molecule followed by (2) binding at the extra-helical site. We find that RBFE can be applied to calculate the free energy of each step, allowing the uncoupling of observed binding free energy from the influence of membrane affinity. This protocol may be used to identify binding hot spots at extra-helical sites and guide drug discovery programs toward optimizing intrinsic activity at the target.



INTRODUCTION

Alchemical relative binding free-energy (RBFE) calculations can be achieved by applying methods such as free-energy perturbation (FEP) or thermodynamic integration (TI) to simulations of the system of interest, often performed by molecular dynamics (MD) or Monte Carlo (MC) methods.¹ RBFEs determine the difference in free energy of binding between two structurally related ligands, which is of enormous benefit to the lead optimization stage of drug discovery projects. Although initially coming to a fore during the 1980s,^{2–4} alchemical free-energy methods have only recently been taken up by the pharmaceutical industry, starting with performance gains displayed by Schrodinger's *FEP+* running on GPU cards⁵ followed more recently by a number of competing packages and academic codes that are able to demonstrate similarly predictive results.^{6–9}

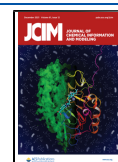
The bulk of reported RBFE studies concern ligands binding to soluble proteins, with achievable performance in the range of <1 kcal/mol mean unsigned error (MUE) compared to the experiment, delivering suitable predictive performance for the optimization of a chemical series during a drug discovery program.⁵ However, fewer studies have investigated the application of RBFE to membrane proteins such as GPCRs, despite the fact that GPCRs constitute ~30% of drug targets.¹⁰ Lenselink *et al.* applied *FEP+* to a number of different ligands across four GPCR targets in a retrospective manner, finding good correlation with measured activity and a root-mean-square error (RMSE) of 0.8 kcal/mol.¹¹ Applying their protocol prospectively on the A_{2A} receptor (A_{2A}R) resulted in the

identification of novel, potent antagonists for this receptor. The A_{2A}R was also used as a validation for the *QLigFEP* package, displaying a similar performance as *FEP+*.¹¹ Deflorian *et al.* undertook an in-depth study on the application of *FEP+* to the A_{2A} and orexin-2 receptors, achieving an MUE of ~0.6 kcal/mol for both targets with a careful system setup, equilibration, and sampling, including initial water placement, amino acid side-chain protonation states, and ligand stability considerations.¹² These reports and others suggest that RBFE calculations may successfully be applied to the orthosteric site of GPCRs during a drug discovery project.

The process of ligand binding to a membrane-embedded target is complicated by the presence of the membrane compartment—affinity of the ligand to the membrane alone may confound experimentally determined binding to the receptor. Specifically, the microkinetic model suggests that a drug may accumulate in the membrane, such that the membrane compartment acts as a depot for the drug.^{13–16} Upon membrane partitioning, the reduction in dimensionality may increase the collision rate of the drug with the target. Finally, re-binding of the drug to the same or a nearby receptor may be promoted.^{17,18}

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Due to the capture of the drug in the membrane depot, the enhanced local concentration increases the observed association to bind the receptor and therefore the observed binding affinity since the observed association rate constant (k_a) is dependent on the ligand concentration:

$$k_a = [L] \cdot k_{on} + k_{off} \quad (1)$$

Measurements of ligand binding at membrane-embedded receptors may therefore be influenced by an increase in the local ligand concentration around the receptor induced by affinity of ligands for the membrane compartment since k_{on} is concentration dependent. Indeed, Sykes *et al.* observed a direct correlation between k_{on} and K_{mem} (a measurement of membrane binding) for a collection of β 2AR ligands yet no such relationship with k_{off} , which is independent of the ligand concentration and proposes a “true” or “intrinsic” affinity of a ligand to a receptor (K_{int}) as^{14,19,20}

$$pK_{int} = pK_{obs} - pK_{mem} \quad (2)$$

where K_{obs} is the measured dissociation constant to bind the receptor and K_{mem} is the membrane partition coefficient. By using pK_{int} , affinities for the receptor are corrected for membrane binding and used the solution phase as reference, thus allowing easier interpretation of structure–activity relationships (SAR) and ligand optimization.¹⁵

These considerations are particularly important when studying the binding of drugs to a lipid-exposed site.^{21,22} By definition, ligands must partition into the membrane to access the binding site and therefore likely have high affinity for the membrane alone. The microkinetic model allows interpretation of ligand-binding at lipid exposed sites (see Figure 1), which can split into

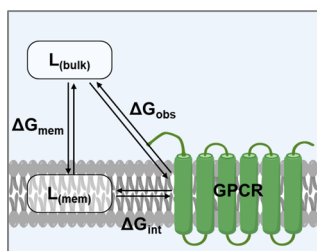


Figure 1. Ligand binding to a lipid-exposed site with observed affinity, ΔG_{obs} , can be uncoupled into two steps: membrane partitioning (ΔG_{mem}) and association to the lipid-exposed binding site (ΔG_{int}).

(1) partitioning of the drug into the membrane (ΔG_{mem}) followed by (2) transfer from the membrane compartment to the ligand binding site (ΔG_{int}). These steps are combined into a single experimental observation of binding, ΔG_{obs} . Upon inspection of Figure 1 and eq 2, the “true” or “intrinsic” affinity of a ligand at a receptor may be determined as

$$\Delta G_{int} = \Delta G_{obs} - \Delta G_{mem} \quad (3)$$

Although drug compounds for membrane-exposed sites must have suitable lipid solubility to access the binding site, drug discovery campaigns must focus on optimizing intrinsic affinity ΔG_{int} to achieve specific activity and/or functional efficacy.

BPTU is a potent antagonist for the P2Y1 purinergic receptor, a class A 7-transmembrane GPCR, discovered by Bristol-Myers Squibb (BMS) for treatment of thrombosis during a drug discovery campaign that was not structurally enabled.²³ The molecular structure of BPTU is shown in Figure 2; structure–activity relationships collected by BMS indicate that a greasy

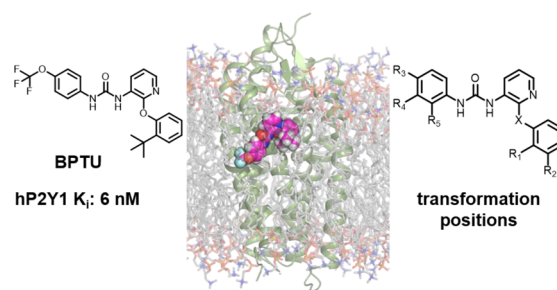


Figure 2. Molecular structure of P2Y1 antagonist BPTU and the location of the membrane-embedded binding site on the extra-helical bundle, plus indication of the substituent and linker positions transformed during relative binding free-energy calculations.

substituent is required at R_1 or R_2 for activity and that binding can be further enhanced by groups at R_4 , R_5 , and, in particular, R_3 (see Figure S1 for a full summary). Subsequent disclosure of the crystal structure of the BPTU-bound P2Y1 receptor by Zhang *et al.* revealed that BPTU in fact bound at an allosteric, membrane-embedded extra-helical site, spanning helices I–III and ECL1.²⁴ Such a membrane-exposed site, coupled with the high lipophilicity of BPTU (clog P , 6.71), suggests that BPTU first partitions into the membrane before transferring to the antagonist site. This binding pathway is supported by a study from Yuan *et al.* using metadynamics to investigate the binding process of BPTU to the allosteric site on the P2Y1 receptor, finding that the lowest energy pathway involves first partitioning into the membrane before undergoing 2D diffusion to associate with the extra-helical ligand binding site.²⁵

In this work, we study the performance of RBFE to predict the affinity of a congeneric antagonist series to bind the membrane-exposed site on the extra-helical bundle of the P2Y1 receptor. We find that relative binding free energies can be calculated with a similar error as reported for soluble targets, with an MUE of 1.08 kcal/mol. By modifying the thermodynamic cycle, we next calculate ΔG_{mem} and ΔG_{int} for two sub-series of antagonists, finding that ΔG_{int} reveals binding hot spots at the antagonist site. This protocol allows uncoupling of membrane affinity ΔG_{mem} and intrinsic affinity for the site ΔG_{int} from the observed affinity ΔG_{obs} , and may assist in future applications of RBFE to optimization of drugs targeting membrane-embedded binding sites.

METHODS

The data set used in this work consists of 30 antagonists with measurable K_i at the P2Y1 receptor reported with the discovery of BPTU (see Figure S1).²³ Binding data reported by Chao *et al.* were determined using a radioligand binding assay, measuring the inhibition of [β -33P]-2MeS-ADP binding from cloned human P2Y1 receptors expressed in HEK293 cells. Inhibition constants K_i were converted into free energies of binding via $\Delta G = -RT \ln(K_i)$, with $T = 298.15$ K. The crystal structure of the P2Y1–BPTU complex (4xnv; 2.2 Å resolution)²⁴ was downloaded from the Orientations of Proteins in Membranes (OPM) database.²⁶ The remaining 29 antagonists were docked to the P2Y1 receptor using Schrodinger *GlideSP* and core restraints with BPTU as the reference ligand coordinates.^{27,28}

To prepare the system for simulations, the receptor was processed with CCG MOE protein preparation, assigning standard amino acid protonation states at pH 7.4, and optimization of the hydrogen bonding network.²⁹ Missing residues in ICL3 were not filled; rather, termini were capped

with ACE and NME capping groups. Waters within the helical bundle were placed using 3D-RISM.^{30,31} A POPC lipid bilayer was then built around the P2Y1 receptor using the *PACKMOL-Memgen* utility,³² specifying a minimum distance of 10 Å between the receptor and box edge and a water layer thickness of 15 Å with NaCl salt concentration of 150 mM. This resulted in 149 lipid molecules, 10,503 water molecules, 7 sodium cations, and 27 chloride anions. A bilayer-only system was also constructed using an in-house membrane builder, consisting of 72 POPC molecules and 2786 waters. BPTU was placed on the water layer above this membrane-only system, at a distance of 35 Å from the bilayer center for membrane partitioning simulations.³³

The ternary complex of a P2Y1–BPTU–POPC lipid bilayer was first equilibrated for 100 ns using *Amber20* and *PMEMD CUDA*.^{34–37} The P2Y1 receptor was modelled with ff19SB,³⁸ water was represented by TIP3P,³⁹ POPC lipids with Lipid14,^{40–42} counter ions with JC parameters,⁴³ and antagonists with parm@Frosst,⁴⁴ a small-molecule force field that extends ff99SB⁴⁵ with conformationally averaged AM1-BCC charges.^{46–49} The system was minimized for 1000 steps using *sander*, with the first 500 steps using the steepest descent method and the remaining steps using the conjugate gradient method.⁵⁰ A longer minimization step was then performed with *PMEMD CUDA* of 10,000 steps; the first 5000 steps used the steepest descent method and the remaining steps used the conjugate gradient method. Next, the system was heated from 0 to 100 K in a 5 ps constant volume run using Langevin dynamics,⁵¹ with restraints of 10 kcal/mol/Å² applied to all receptor, drug, and lipid atoms. The volume was then allowed to change freely, and temperature increased to 298 K with a Langevin collision frequency of $\gamma = 1 \text{ ps}^{-1}$, with anisotropic Berendsen control of the pressure around 1 atm applied by coupling the periodic box with a time constant of 2 ps for 100 ps.⁵² The system was then run in the NPT ensemble with semi-isotropic Berendsen pressure coupling, with the X- and Y-dimensions coupled and the Z-dimension allowed to change freely for 1 ns, with restraints on 5 kcal/mol/Å² applied to receptor backbone atoms and drug atoms only. An additional 1 ns of NPT simulation followed, with 5 kcal/mol/Å² restraints on receptor carbon-alpha and drug atoms only. Finally, all restraints were removed and the system was simulated in the NPT ensemble for 100 ns. Partitioning simulations were also performed using the BPTU-bilayer-only systems with 72 POPC lipids, in an identical manner to previous work.³³ Three repeat simulations were performed, starting with the BPTU drug in the water phase above the POPC bilayer, each of 200 ns in length. In all cases, the BPTU drug transitioned into the membrane phase within 20–50 ns.

With P2Y1–BPTU–POPC bilayer and BPTU-bilayer-only systems equilibrated, it was then possible to transition into relative binding free-energy calculations. To plan these calculations, a perturbation map is required, as may be obtained from programs such as *LOMAP*.⁵³ In this work, we used a similar approach to *LOMAP*, but rather than a 2D similarity score, we use an ROCS 3D TanimotoCombo score to weight the edges, which takes into account the molecular shape and electrostatics.⁵⁴ Inclusion of 3D information allows the avoidance of ambiguities about the correct maximum common substructure that may arise when using 2D information alone.⁷ The 30 antagonists all share a common substructure and are aligned in 3D after the *GlideSP* core restrained docking. Next, the ROCS TanimotoCombo score is calculated in place for every ligand

against every other ligand and a graph is assembled using *NetworkX*,⁵⁵ with each molecule as a node and the ROCS 3D similarity score as the edge weight. The edges are sorted by weight, from a low to high 3D similarity score. The graph is then iteratively pruned of edges, with the requirement that it remains a fully connected graph and that all molecules remain in a cycle size of four or less. Finally, the two most distal cycles are connected via a ligand pair with the highest 3D similarity score. The perturbation map derived from this process for the 30 P2Y1 antagonists is shown in Figure S2, and a TanimotoCombo similarity matrix for the full set is given in Figure S3, with ligand pairs selected for perturbation indicated. For calculations of ΔG_{mem} and ΔG_{int} , two sub-series were identified around ligand 11a (transforming to four related analogs) and ligand 6a (transforming to eight related analogs), in both cases, using a star map and the thermodynamic cycles are shown in Figure 3.

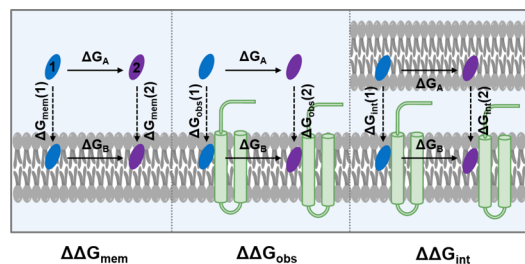


Figure 3. Thermodynamic cycle used to calculate $\Delta\Delta G_{\text{mem}}$ ($\Delta G_{\text{B}} - \Delta G_{\text{A}}$) between example ligands 1 & 2, $\Delta\Delta G_{\text{obs}}$ ($\Delta G_{\text{B}} - \Delta G_{\text{A}}$) between example ligands 1 & 2, and $\Delta\Delta G_{\text{int}}$ ($\Delta G_{\text{B}} - \Delta G_{\text{A}}$) between example ligands 1 & 2. In all cases, the transformations with bold arrows were those computed.

Free-energy simulations were then performed using *Amber20* thermodynamic integration (TI);^{7,34,56,57} all simulations followed an identical procedure and simulation settings. For the calculation of ΔG_{obs} , ligands were transformed at the membrane-embedded antagonist site and in the water phase; for the calculation of ΔG_{mem} , ligands were transformed in the bilayer-only phase and water phase; for the calculation of ΔG_{int} , ligands were transformed at the membrane-embedded antagonist site and in the bilayer-only phase (see Figure 3). Atoms were classified as being “alchemical” by comparing Cartesian coordinates, Amber atom types, and bond connectivity using *RDKit*.⁵⁸ The “unified” protocol was used, transforming van der Waals and electrostatics in one step, with application of second-order smooth-step soft-core potentials.⁵⁹ To initiate calculations, systems were first minimized for 1000 steps, with the first 500 steps using the steepest descent method and the remaining steps using the conjugate gradient method with a lambda value $\lambda = 0.5$ and restraints of 5 kcal/mol/Å² applied to all atoms other than water and salt molecules. The system was then heated to 298 K in a 20 ps NVT simulation using Langevin dynamics, again with a lambda value $\lambda = 0.5$ and restraints of 5 kcal/mol/Å² on all atoms other than water and salt molecules. To relax the system box size, the system was then simulated for 100 ps in the NPT ensemble using the Monte Carlo barostat, maintaining restraints and the lambda value $\lambda = 0.5$. The *SHAKE* algorithm was applied to constrain all bonds involving hydrogen,⁶⁰ with the exception of the alchemical ligands for which *SHAKE* was turned off using the “tishake” keyword. Each stage used a 1 fs time step and a cutoff value of 9 Å, beyond which PME was used to treat electrostatics and an analytical dispersion correction was

applied to van der Waals interactions.⁶¹ Following this, 12 parallel simulations were launched for each of the two legs of the transformation with lambda values set to allow Gaussian quadrature integration. For each lambda window, the system was again minimized and heated using the identical protocol mentioned previously. All restraints were then removed and the system was moved into an NPT equilibration period of 5 ns. A production simulation of 5 ns NPT then followed and dV/dL data were collected. For each transformation; two repeats were performed and final free-energy values are reported as the average and standard deviation across the two repeats, with free energies calculated via integration of dV/dL curves using Gaussian quadrature integration. Final statistical data analysis was performed with the *arsenic* package to determine pairwise $\Delta\Delta G$ correlations and transform predictions to compound ΔG values via the perturbation map and a maximum likelihood estimator.⁶²

Amber TI has been validated in a number of benchmarking studies.^{6,7,63} Prior to application to the P2Y1 system, an in-house validation was performed using this setup (ROCS 3D perturbation map, parm@Frosst small-molecule force field, Amber20 TI) on the well-studied dataset of eight benchmark systems compiled by Schrodinger.⁵ Commensurate performance was found in *FEP+* and *Cresset Flare*,⁹ indicating that this protocol is suitable for RBFE calculations on drug discovery projects. Full results are provided in Table S1.

To compare $\Delta\Delta G_{\text{mem}}$ values to the difference in compound lipophilicity, calculated log *P* (clog *P*) values were determined using the *RDKit* from the canonical isomeric SMILES of each molecule.⁵⁸

Finally, for analysis of protein–ligand interactions, the *getcontacts* application was run on select protein–ligand trajectories from the 11a antagonist subset.⁶⁴ All van der Waals and hydrogen bonding contacts were computed. The difference (delta) between each contact was determined, comparing each ligand in the subset to the result for antagonist 11a.

RESULTS AND DISCUSSION

RBFE Can Rank–Order Antagonists at a Lipid-Exposed Site. The RBFE results for $\Delta\Delta G_{\text{obs}}$ of the full set of 30 antagonists are shown in Figure 4, in which the ligands are transformed in the water phase and while bound at the extra-helical site of the P2Y1 receptor. The pairwise $\Delta\Delta G$ values are calculated with an MUE of 1.08 kcal/mol in comparison to the experiment, with all cycle closure hysteresis values being <1

kcal/mol (see Table S2). Transforming to compound ΔG values results in an MUE of 1.29 kcal/mol with a correlation coefficient of $r^2 = 0.58$ and Spearman's $\rho = 0.76$. This data indicates that RBFE can be applied to ligand binding at membrane-exposed sites to prioritize P2Y1 antagonists for observed binding (ΔG_{obs}). We also evaluated the null models (clog *P*) and molecular weight (MW) for prediction of binding affinity, finding no correlation between ΔG_{obs} and MW ($r^2 = 0.03$) and a trend with clog *P* ($r^2 = 0.39$) as may be expected for a hydrophobic, membrane-exposed binding site (see Figure S4).

Uncoupling Membrane and Intrinsic Components from Observed Binding. Next, the thermodynamic cycles shown in Figure 3 were used to determine $\Delta\Delta G_{\text{obs}}$, $\Delta\Delta G_{\text{int}}$, and $\Delta\Delta G_{\text{mem}}$ for two sub-series of antagonists, using star maps to transform ligand 11a to four related analogs and ligand 6a to eight related analogs. A closure term was also calculated as $\Delta\Delta G_{\text{obs}} - \Delta\Delta G_{\text{int}} - \Delta\Delta G_{\text{mem}}$, which should approach zero provided the consistency across the full cycle of water → membrane → lipid-exposed site. The results are given in Tables 1 and 2. Regarding the overall performance, the agreement between calculated $\Delta\Delta G_{\text{obs}}$ and the experiment was very similar to the full set, with an MUE of 1.03 kcal/mol (see Figure S5) across the 12 star map transformations.

Upon inspection of Table 1, there is a good agreement between experimental and calculated $\Delta\Delta G_{\text{obs}}$ terms, with the exception of the 11a → 11b transformation, which is over-predicted. This transformation also had the largest closure term amongst the set. However, the rank ordering by calculated $\Delta\Delta G_{\text{obs}}$ matches the experiment and a number of values are quantitatively very close. The transformations in Table 1 are ordered by $\Delta\Delta G_{\text{int}}$, which is the calculated affinity for antagonists to bind to the extra-helical site after removing membrane binding. This re-ordering of antagonist affinity then prioritizes ligands 11b and 11f, which both place greasy substituents (*tert*-butyl and tri-fluoromethyl, respectively) at position R1, over ligands 11c and 1, which place identical groups at position R2. This indicates that groups at R1 interact with a hydrophobic hot spot on the P2Y1 receptor, analysis of ligand contacts (see Figure S6) indeed shows a large increase in van der Waals contacts with residue Leu102 for ligands 11b and 11f versus 11a, whereas 11c and 1 do not show a similar change in van der Waals' contact. These results indicate that although *tert*-butyl at either R1 or R2 position improves $\Delta\Delta G_{\text{obs}}$, only the R1 position shifts $\Delta\Delta G_{\text{int}}$ whereas the R2 position acts to boost membrane affinity only. Substituents at R1 are found to noticeably boost $\Delta\Delta G_{\text{int}}$ via interaction with a hydrophobic hot spot at Leu102 (see Figure 5). Such information may allow drug discovery campaigns to target affinity hot spots, thereby increasing specific activity at the target, rather than membrane affinity alone. This is a similar approach employed by the lipophilic efficiency (lipE) metric,^{65–67} which allows drug discovery projects to track improvements in specific binding at the target as opposed to increasing lipophilicity alone. Lipophilicity of a drug candidate has a strong influence on potency, pharmacokinetics, and toxicity and is thus a well-documented property to control in order to increase the likelihood of success.⁶⁸ Increasing lipophilicity typically results in higher off-target binding, poorer solubility, and increased metabolic clearance; drug discovery campaigns should therefore track gains in specific binding as opposed to lipophilicity-induced association with the target.

The same protocol was then run on a second subset of antagonists around ligand 6a, which places different substituents

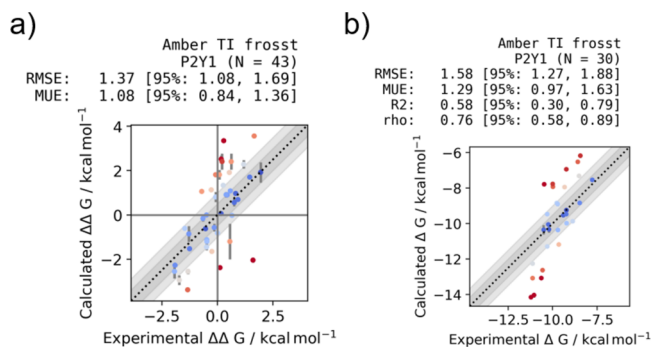
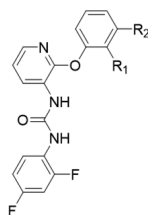


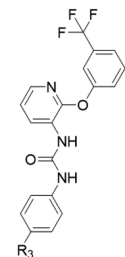
Figure 4. (a, b) Results for 43 pairwise transformations ($\Delta\Delta G_{\text{obs}}$) and transforming to 30 compound values (ΔG_{obs}) using Amber20 TI on P2Y1 antagonists binding at a lipid-exposed site.

Table 1. Results for Calculations of $\Delta\Delta G_{\text{obs}}$, $\Delta\Delta G_{\text{int}}$, and $\Delta\Delta G_{\text{mem}}$ for a Subset of Antagonists around Ligand 11a Using a Star Map

Transform	R1	R2	exp $\Delta\Delta G_{\text{obs}}$ (kcal/mol)	calc $\Delta\Delta G_{\text{obs}}$ (kcal/mol)	calc $\Delta\Delta G_{\text{int}}$ (kcal/mol)	calc $\Delta\Delta G_{\text{mem}}$ (kcal/mol)	Closure ($\Delta\Delta G_{\text{obs}}$ - $\Delta\Delta G_{\text{int}}$ - $\Delta\Delta G_{\text{mem}}$)
11a-> 11b	H-> C(CH ₃) ₃	H	-2.79	-4.93 ± 0.03	-3.34 ± 0.50	-2.67 ± 0.08	1.08
11a-> 11f	H-> CF ₃	H	-1.45	-1.71 ± 0.24	-0.33 ± 0.59	-1.07 ± 0.43	-0.32
11a-> 11c	H	H-> C(CH ₃) ₃	-2.43	-2.39 ± 0.44	-0.07 ± 0.78	-2.32 ± 0.53	0.01
11a-> 1	H	H-> CF ₃	-1.95	-2.00 ± 0.001	0.37 ± 0.81	-2.16 ± 0.18	-0.21

**Table 2. Results for Calculations of $\Delta\Delta G_{\text{obs}}$, $\Delta\Delta G_{\text{int}}$, and $\Delta\Delta G_{\text{mem}}$ for a Subset of Antagonists around Ligand 6a Using a Star Map**

Transform	R3	exp $\Delta\Delta G_{\text{obs}}$ (kcal/mol)	calc $\Delta\Delta G_{\text{obs}}$ (kcal/mol)	calc $\Delta\Delta G_{\text{int}}$ (kcal/mol)	calc $\Delta\Delta G_{\text{mem}}$ (kcal/mol)	Closure ($\Delta\Delta G_{\text{obs}}$ - $\Delta\Delta G_{\text{int}}$ - $\Delta\Delta G_{\text{mem}}$)
6a-> 6i	H-> C(CH ₃) ₃	-1.01	-3.08 ± 0.19	-0.86 ± 0.14	-1.81 ± 0.14	-0.40
6a-> 6f	H-> F	-0.10	-1.14 ± 0.002	-0.39 ± 0.01	-1.11 ± 0.09	0.35
6a-> 6h	H-> CH ₃	-0.75	-1.14 ± 0.07	-0.34 ± 0.16	-0.49 ± 0.22	-0.31
6a-> 6m	H-> N(CH ₃) ₂	0.64	0.44 ± 0.10	-0.12 ± 0.61	-0.60 ± 0.55	1.16
6a-> 6g	H-> Cl	-0.54	-1.86 ± 0.06	-0.08 ± 0.17	-1.52 ± 0.23	-0.27
6a-> 6l	H-> OCH ₃	-1.24	-0.90 ± 0.001	0.33 ± 0.06	-0.35 ± 0.39	-0.88
6a-> 6j	H-> OCF ₃	0.36	-3.03 ± 0.01	0.50 ± 0.27	-3.64 ± 0.12	0.11
6a-> 6n	H-> CCN	0.71	-0.34 ± 0.004	0.95 ± 0.37	-1.22 ± 0.36	-0.07



on the terminal aryl ring (position R3), see Table 2. In general, calculated $\Delta\Delta G_{\text{obs}}$ rank orders are in agreement with experimental $\Delta\Delta G_{\text{obs}}$; however, there are some discrepancies, in particular, ligands 6i and 6j are over-predicted. This may be due to small-molecule or lipid force fields failing to correctly capture free energies of membrane partitioning; it has been proposed to use different sets of implicitly polarized charges for polar and non-polar phases to better model drug–membrane interactions.⁶⁹ Furthermore, although the P2Y1 antagonists are neutral molecules, constant-pH MD has been shown to capture the pH-coupled partitioning of charged compounds.⁷⁰

$\Delta\Delta G_{\text{int}}$ values trend with experimental/calculated $\Delta\Delta G_{\text{obs}}$, indicating that the contribution of membrane binding may be alike across this subset of antagonists (see Figure S7 for a plot of experimental $\Delta\Delta G_{\text{obs}}$ vs calculated $\Delta\Delta G_{\text{int}}$ for the 6a and 11a subsets). Indeed, $\Delta\Delta G_{\text{mem}}$ is found to be relatively similar for each of the small substituents, with the exception of large predictions for 6i and 6j. We also found that despite the small

dynamic range, there is a good correlation between $\Delta\Delta G_{\text{mem}}$ and the difference in clog *P* (see Figure S8), with a correlation coefficient of $r^2 = 0.48$. The subset 6j is a large outlier from this trend, again suggesting that $\Delta\Delta G_{\text{mem}}$ is over-predicted for this molecule. The R3 position is oriented toward the face of the aromatic ring of Phe66, potentially explaining the preference for small, hydrophobic substituents upon re-ordering by calculated $\Delta\Delta G_{\text{int}}$ such as fluoro (6f) and methyl (6h).

CONCLUSIONS

In this work, we first apply RBFE to calculate $\Delta\Delta G_{\text{obs}}$ for 30 P2Y1 antagonists binding at a lipid-exposed extra-helical site, finding that pairwise $\Delta\Delta G$ values are calculated with an MUE of 1.08 kcal/mol in comparison to the experiment, indicating that RBFE may be applied in a predictive manner to drug discovery targets with membrane-embedded binding sites. Next, to uncouple intrinsic affinity for the extra-helical site from membrane binding, we use different thermodynamic cycles to

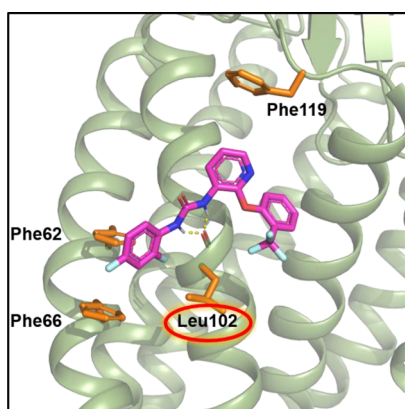


Figure 5. Features of the P2Y1 antagonist lipid-exposed binding site, with antagonist 11f bound. All antagonists make hydrogen bonding interactions to the backbone carbonyl of Leu102. Calculation of $\Delta\Delta G_{\text{int}}$ suggests that the Leu102 side chain is a hydrophobic hot spot.

calculate $\Delta\Delta G_{\text{int}}$ and $\Delta\Delta G_{\text{mem}}$ for two subsets of antagonists. We find that $\Delta\Delta G_{\text{int}}$ re-orders the SAR and uncovers a hydrophobic hot spot at the P2Y1 binding site. Such a protocol may allow drug discovery programs to optimize the intrinsic affinity and hence specific activity, as opposed to optimizing the membrane affinity alone. This is a similar strategy employed by the lipophilic efficiency metric, to optimize specific binding at the target as opposed to increasing lipophilicity, which may be important to retain suitable ADME properties, such as solubility and clearance. We propose that the protocol used in this work may be utilized for such an optimization approach during drug discovery campaigns targeting a membrane-embedded binding site.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.1c01147>.

Molecular structures for the 30 antagonists, perturbation map, Amber20 TI frosst results for eight benchmark systems, null model results, getcontacts plots for the 11a subset, $\Delta\Delta G_{\text{obs}}$ versus $\Delta\Delta G_{\text{int}}$ and $\Delta\Delta G_{\text{mem}}$ for 11a and 6a subsets, and the $\Delta\Delta G_{\text{mem}}$ trend with delta clog *P* (PDF)

Full table for the 30 antagonists with IDs, SMILES strings, experimental hP2Y1 Ki values, and all calculated free energies (XLSX)

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Notes

The authors declare no competing financial interest.

Simulations were performed with Amber20 (<http://ambermd.org/index.php>) and related force fields. Protein preparation was performed with CCG MOE (<https://www.chemcomp.com/>). AM1-BCC charges were calculated with Openeye QuacPac (<https://www.eyesopen.com/quacpac>). RDKit is available under BSD 3 license (<https://github.com/rdkit/rdkit>). GetContacts is available under Apache 2 license (<https://github.com/getcontacts/getcontacts>). A full table for the 30 antagonists with IDs, SMILES strings, experimental hP2Y1 Ki values, and all calculated free energies is available as an Excel file in the Supporting Information. Topology and coordinate files are available upon request.

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■ ABBREVIATIONS

MD, molecular dynamics; TI, thermodynamic integration; FEP, free-energy perturbation; RBFE, relative binding free energy; GPCR, G-protein coupled receptor

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